

Low-Fi VR Controller: Bringing 6DOF Interaction to Mobile VR

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ABSTRACT

As virtual reality (VR) becomes an everyday technology, it is important to ensure it remains broadly accessible and affordable. Mobile VR is low-cost and accessible to everyone with a smartphone. However, it does not support a 6 degrees of freedom (6DOF) input device nor an effective way to select objects using the common techniques employed in commercial VR (i.e., ray-casting and virtual hand). Our Low-Fi VR Controller prototype provides a paper-based 6DOF input device for mobile VR. The controller's pose is tracked by the smartphone's camera and users can select objects by pointing the controller to direct a raycast or use the controller like a virtual hand. Selection indication can be performed using 300 ms dwell or a manual technique we refer to as Marker Selection comparable to click activation. This demo allows users to test the controller using both techniques and selection indication methods while performing ISO-9241-411 selection tasks. The top scores throughout the demo session will be displayed in leaderboards to promote friendly competition among attendees.

Keywords: Virtual reality, pointing device, mobile VR.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction devices

1 INTRODUCTION

Tech giants, such as Meta, Apple, and Sony, are consistently making advancements in commercial virtual reality (VR) displays and have positioned VR to become an everyday technology for work, online social spaces, and entertainment. However, VR's high cost prevents users of lower socioeconomic statuses from adopting the technology, leaving them excluded and increasing the digital divide.

Mobile VR (MVR) offers a low-cost alternative to modern VR head-mounted displays (HMDs). A smartphone is inserted into a cardboard viewer and serves as both the processor and display. Ranging in price from \$10-20 USD, MVR has the potential to make VR more affordable and broadly accessible. However, unlike modern VR devices, MVR supports only 3 degrees of freedom (DOF) head rotation, head-gaze based selection, and input via a button on the cardboard shell. It does not support 6DOF tracked handheld devices required for the most common VR object selection techniques: controller-based ray-casting and virtual hands [4]. Due to these limited input capabilities, MVR cannot support most VR applications that require interaction such as locomotion, or object selection and manipulation. Thus, it generally cannot provide the full range of benefits of VR in training, education, health applications, collaborative work, and entertainment [2].

Past solutions to improving MVR selection have either yielded measurably worse solutions compared to commercial VR or

provided 3 or 6DOF interaction only through the use of external expensive devices [2]. Proposed low-cost solutions involved HMD extensions or gestures on the HMD surface, but only supported UI menu interaction [1]. Other solutions used camera-based tracking and hand gestures to support virtual hand and ray-based selection [4]; however, the hand tracking and gesture recognition software was unreliable. Other researchers have used inertial sensors of a smartwatch or second smartphone for 3 or 6DOF ray-based selection [3, 5]. However, this extra cost does not maintain MVR's affordability.

The absence of affordable 6DOF MVR interaction motivated the development of our Low-Fi VR Controller [2], a low-cost 6DOF tracked controller to support ray-casting and virtual hand selection. The controller only requires the smartphone camera, paper, and cardboard. To provide a method for manual selection indication (i.e., a "click"), users hold a second cardboard piece in the other hand. See section 2 for full details. The Low-Fi VR Controller can help expand the types of interactions and applications mobile VR can support.

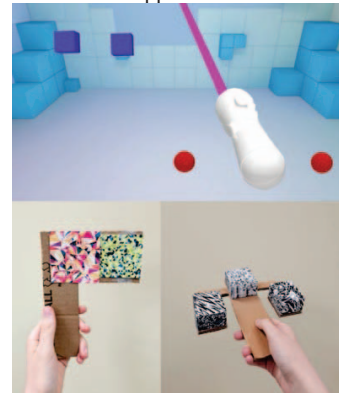


Figure 1: Low-Fi VR Controller and virtual representation.

2 CONTROLLER DESIGN

Full details of our design process are documented in our previous work [2]. To support the low-fidelity and accessible nature of MVR, we ensured all software and materials used for the controller remained in the price range of a Google Cardboard (\$10-20 USD) and was portable. The controller is made with solely paper-based materials commonly found in homes.

We designed a camera-based tracking system in Unity 2021. We used the Google XR Plugin v.1.2 to support stereoscopic rendering and Vuforia v.10.15.3, an augmented reality (AR) SDK that tracks the pose of image markers relative to the smartphone.

A unique feature of Vuforia not available in other AR SDKs (e.g., ARCore) is its multi-target feature: the ability to track multiple images forming one object. Our controller used three cube-shaped multi-targets. Each multi-target has a unique image on each side of the cube, allowing Vuforia to determine its pose. We created the images in Adobe Photoshop following Vuforia's best practices for choosing target images (e.g., high contrast, non-repeating patterns, non-organic shapes). To assemble the multi-targets, we first printed them on paper, then folded and fastened them with tape. The controller handle is made of cardboard.

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The three multi-targets form an isosceles triangle (see Figure 2). We chose this configuration so the controller remained visible while being held farther to the left, right, and forward. This artificially increases the tracking boundaries compared to just using a single central marker, alleviating a common issue with MVR optical tracking: a small effective tracking area [4, 5].

The virtual representation of the controller (a 3D model of a PlayStation VR controller¹) is positioned in the centre of the markers. To initially obtain its position, all three multi-targets must be tracked simultaneously in a frame. After this, since the distances between the multi-targets are known, only one tracked multi-target is needed to calculate the position. The controller's orientation is determined by the average quaternion of the currently tracked multi-targets in a given frame. If all three multi-targets lose tracking, the virtual controller freezes in the last known position and turns red to notify the user.

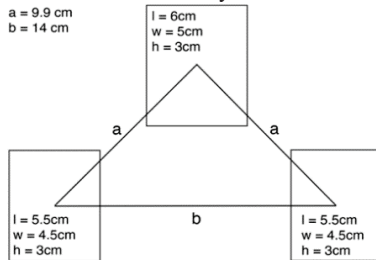


Figure 2: Multi-target configuration and measurements.

2.1 Supported Selection Techniques

Our previous experiment measured selection performance only with ray-casting and dwell-based selection indication [2]. However, the Low-Fi VR controller can support both ray-based and virtual hand selection [4]. For ray-based, a ray originates from the virtual controller's position, and its direction is determined by the controller's tracked orientation. For the virtual hand technique, (i.e., direct touch), the virtual controller has a small cursor at its tip and selections are performed by touching objects directly.

The controller currently supports two methods of selection indication. The first is dwell where the ray or controller must touch the object for 300 ms, after which selection occurs. The second technique is Marker Selection, which uses two tracked images attached to a second piece of cardboard held in the user's non-dominant hand (see Figure 1). The user must keep at least one of the tracked markers in view of the camera. To confirm selection, the user can simply hide both markers from the camera by turning or swiping the markers away from the camera. This method for selection indication provides more accuracy and control over where and when a selection occurs.

3 DEMO WALKTHROUGH

During the demo, attendees can test the controller using two selection techniques. The opening scene (see Figure 3) includes instructions and four leaderboards, detailed below. Using the controller to select start, participants will be taken to another scene where the demo will begin. Participants have the option to test the controller in a series of six ISO-9241-411 reciprocal selection tasks using the ray-based selection technique with Marker Selection (see Figure 4 left) or the virtual hand technique with dwell selection indication (see Figure 4 right). After completing a trial, three performance metrics are displayed to the user: their fastest selection time, average selection time, and their

¹The controller model is available at <https://sketchfab.com/3d-models/playstation-vr-645bb34486a54e618247c51c98f56205>

throughput score. At any time between ISO trials, the participant can switch between ray-casting and virtual hand or choose to go back to the main scene with the leaderboards. However, participants must complete all six ISO tasks consecutively for a given selection technique to qualify for a leaderboard.

We implemented four leaderboards to gamify the demo experience. The leaderboards save scores from participants throughout the demo session to encourage friendly competition between attendees. Two leaderboards are for the ray technique, while the two others are for the virtual hand technique. For each technique, one leaderboard displays the five best throughput scores participants achieved out of any of their six ISO trials. The second leaderboard, one for each technique, displays the five best average selection times of a given trial. When a participant finishes their sixth ISO trial, they will be notified whether they made a leaderboard or not. If they did, they are prompted to enter three initials that will be displayed beside their score on the leaderboard.



Figure 3: Start scene with four leaderboards.

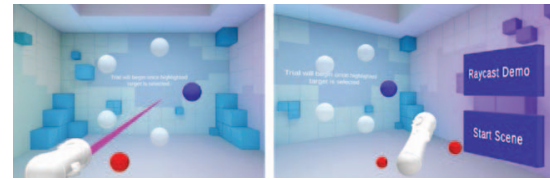


Figure 4: Controller using raycast (left) and virtual hand (right) techniques in ISO-9241-411 selection task.

4 RESEARCH DIRECTIONS

In future work, we plan to iterate on the design of the controller and system to improve its tracking accuracy and usability. Two major goals we have are designing a one-handed version of the controller with manual selection indication and validating the control for use with other fundamental VR tasks, namely travel and object manipulation.

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